Euclid Book I: Reconstruction
Proposition I.5
In an Isosceles Triangle the Base Angles are Congruent

CGJteamLab

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1 Introduction

Proposition I.5 is the first theorem of Book I devoted specifically to the geometry of isosceles triangles. It establishes that the angles at the base of an isosceles triangle are congruent. Moreover, if the equal sides are extended beyond the base, the exterior angles formed by these extensions are also congruent.

In the Elements this proposition is one of the best known results of classical geometry and later became known under the traditional name *pons asinorum*. Although the theorem itself is elementary, it marks the beginning of a systematic study of congruence properties of triangles.

Euclid proves the proposition by combining the congruence theorem of Proposition I.4 with auxiliary constructions on the extensions of the equal sides.

In the present reconstruction the logical organization is different. The theorem is obtained by composing two previously established results of the Hilbert interface. The congruence of the base angles is provided by the theorem on isosceles triangles, while the congruence of the exterior angles follows from Hilbert's theorem on adjacent angles.

Consequently, Proposition I.5 is the first example in which a Euclidean proposition is reconstructed by combining several independent interface theorems rather than by introducing new geometric techniques.

2 The Classical Configuration

Euclid's proof of Proposition I.5 does not use only the original isosceles triangle and the extensions of its equal sides. It introduces additional auxiliary points and joins, producing a larger configuration to which Proposition I.4 can be applied.

Let ABC be an isosceles triangle satisfying

$$AB \cong AC.$$

Produce the equal sides AB and AC beyond the base vertices B and C. Euclid then chooses an auxiliary point on one extension, transfers a corresponding segment to the other extension, and joins the resulting points to the opposite vertices of the base.

The following sequence shows the successive stages of Euclid's configuration.

2.1 Initial Isosceles Triangle

The construction begins with an isosceles triangle ABC satisfying

$$AB \cong AC.$$

At this stage no auxiliary points or segments have yet been introduced.

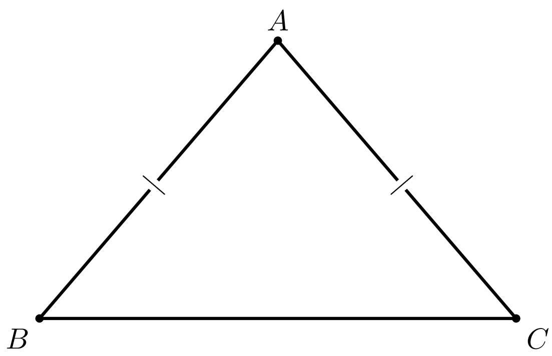

Figure 1: The initial isosceles triangle, with $AB \cong AC$.

2.2 Extension of the Equal Sides

Euclid next produces the equal sides AB and AC beyond the base vertices. Let D lie beyond B on the line AB, and let E lie beyond C on the line AC.

Thus

$$A - B - D, \qquad A - C - E.$$

3 Statement

Let ABC be a nondegenerate isosceles triangle satisfying

$$AB \cong AC.$$

Let the equal sides be extended beyond the base vertices so that

$$A - B - D, \qquad A - C - E.$$

Then

$$\angle ABC \cong \angle ACB,$$

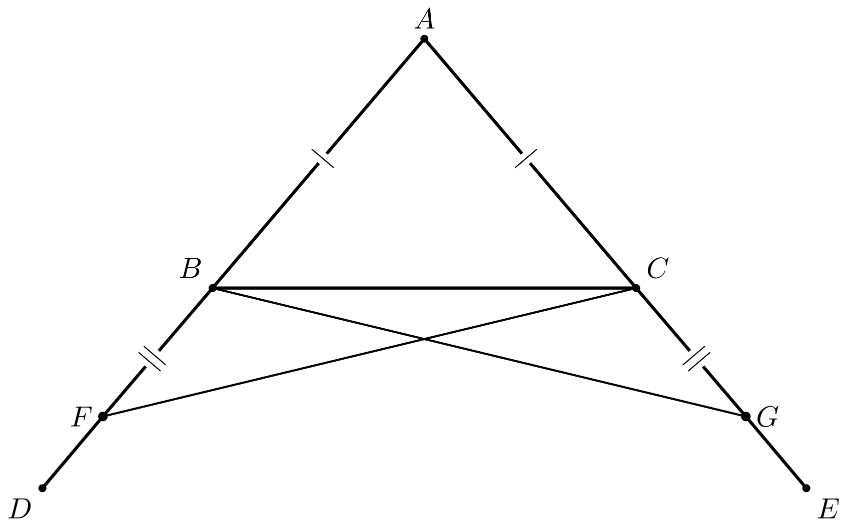

Figure 2: The complete classical configuration for Proposition I.5. The sides are produced beyond B, C; F is chosen on the left extension, G on the right so that $BF \cong CG$; finally FC and GB are joined. The outer points D, E record the rays used in the statement.

and

$$\angle CBD \cong \angle BCE.$$

Thus both the base angles of the isosceles triangle and the exterior angles formed after extending the equal sides are congruent.

Diagrammatic audit. The four successive construction snapshots have been replaced by one complete classical figure. For I.5 the successive states do not represent genuinely different geometric cases: they merely add F, then G, then the two joins. A single well-spaced configuration makes the proof easier to follow.

Wilkins's note confirms that the two genuinely important stages are the two SAS comparisons. First FAC is compared with GAB; this gives, among other consequences, $FC \cong GB$. Then FBC is compared with GCB, which yields the equal angles below the base. Subtracting equal angles then gives the equality of the base angles. These logical stages are described in the proof rather than represented by repeated copies of the same diagram.

4 Classical Proof

Euclid's proof is considerably more elaborate than the modern reconstruction presented later in this chapter. Rather than reasoning directly from the properties of an isosceles triangle, he introduces an auxiliary construction that allows repeated applications of the Side– Angle–Side congruence theorem established in Proposition I.4.

After extending the equal sides of the triangle, an arbitrary point is chosen on one extension. Proposition I.3 is then used to construct an equal segment on the opposite extension. The newly obtained points are joined to the opposite vertices of the base, completing the auxiliary configuration.

The proof proceeds by applying Proposition I.4 twice. The first application establishes congruence relations between the auxiliary triangles and yields equal corresponding sides and angles. These newly obtained equalities provide the hypotheses required for a second application of Proposition I.4.

Finally, Euclid combines the resulting congruence relations with the elementary properties of equal segments and equal angles to conclude that the interior base angles of the original isosceles triangle are congruent. The same argument immediately establishes the congruence of the corresponding exterior angles formed after extending the equal sides.

The auxiliary construction is therefore not an end in itself. Its sole purpose is to transform Proposition I.5 into two successive applications of Proposition I.4, allowing the theorem to be derived using only previously established geometric results.

5 Proof Architecture of the Reconstruction

The formal proof of Proposition I.5 is obtained by composing two previously established theorems of the Hilbert interface.

The first theorem establishes the congruence of the base angles of an isosceles triangle. The second theorem transfers this congruence to the adjacent exterior angles after the equal sides have been extended.

The resulting logical dependency is shown below.

```
Euclid
------
AB ~= AC
   |
produce AB, AC beyond B, C
   |
choose F; I.3 gives G with BF ~= CG
   |
join FC and GB
   |
I.4: SAS on FAC and GAB
   |
I.4: SAS on FBC and GCB
   |
subtract equal angles
   |
base angles equal
and angles below base equal


Hilbert / Lean
--------------
AB ~= AC
   |
hilbert_isosceles_base_angles
   |
angle ABC ~= angle ACB
```

```
   |
A-B-D and A-C-E
   |
hilbert_adjacent_angles_congruent
   |
angle CBD ~= angle BCE
```

Unlike the classical proof of Euclid, the formal development does not repeat the geometric argument establishing the base-angle theorem. Instead, the proposition is reconstructed by composing two independent results already available in the Hilbert interface. Consequently, Proposition I.5 illustrates the modular nature of the library: new Euclidean propositions are obtained by combining previously verified synthetic theorems.

6 Hilbert Reconstruction

At the level of Proposition I.5, the Hilbert proof is much shorter than Euclid's classical construction.

The first conclusion,

$$\angle ABC \cong \angle ACB,$$

is already the theorem

```
hilbert_isosceles_base_angles
```

applied to the noncollinear isosceles triangle ABC.

For the second conclusion, the hypotheses

$$A - B - D, \qquad A - C - E$$

say that the equal sides have been produced beyond the base vertices. The established theorem

```
hilbert_adjacent_angles_congruent
```

transports the congruence of the base angles to their adjacent exterior angles, giving

$$\angle CBD \cong \angle BCE.$$

Thus the formal proposition does not construct Euclid's auxiliary points F, G, and it does not replay the two SAS arguments. Those facts have already been packaged into the Hilbert interface theorems used here.

7 Lean Formalization

The complete implementation is available in the repository. The two interface calls contain the mathematical content of the formal proof.

The base-angle conclusion is immediate:

```
have hBase :=
  hilbert_isosceles_base_angles
    Geo A B C hABC hABAC
```

The exterior-angle conclusion is then obtained from the two extensions and the already established base-angle congruence:

```
have hExterior :=
  hilbert_adjacent_angles_congruent
    Geo
    A B C D
    A C B E
    hABD
    hACE
    hABC
    hACB
    hBase
```

The theorem returns the pair $\langle hBase, hExterior \rangle$. The only additional bookkeeping is the rotated noncollinearity hypothesis required for the second call.

8 Relationship with Earlier Propositions

8.1 Proposition I.2

Euclid's proof requires the transfer of a prescribed segment length to a new position in the auxiliary construction. Proposition I.2 supplies the underlying segment-transport mechanism.

In the Hilbert reconstruction this role is absorbed by the established segment-layoff infrastructure.

8.2 Proposition I.3

After extending the equal sides of the original isosceles triangle, Euclid cuts off an auxiliary segment equal to a prescribed segment. Proposition I.3 provides exactly this operation.

This construction creates the equal-side data needed for the subsequent triangle-congruence arguments.

8.3 Proposition I.4

SAS is the decisive congruence theorem in the classical proof. It is applied to auxiliary triangles in order to propagate the original equality of the two sides into the angle congruences required at the base.

The formal reconstruction uses the corresponding Hilbert SAS theorem and angle-congruence transport.

8.4 Transition to Proposition I.6

Proposition I.5 proves

$$AB \cong AC \Longrightarrow \angle ABC \cong \angle BCA.$$

Proposition I.6 proves the converse implication. Together they establish the first full equivalence between side symmetry and angle symmetry in a triangle.

9 Hilbert and Book Zero Components Used

9.1 Segment Extension and Layoff

The proof extends the equal sides beyond the base vertices and chooses auxiliary points at prescribed distances. These operations are supplied by the established Hilbert order and congruence interfaces.

9.2 SAS

The Hilbert SAS theorem is the central rigidity principle in the reconstruction. It converts the constructed side congruences and the appropriate included-angle congruence into further corresponding side and angle congruences.

9.3 Angle Congruence Transport

The auxiliary triangles naturally produce angles in orientations that do not always coincide syntactically with the final base angles. Symmetry, transitivity, and transport of angle congruence are therefore part of the formal proof rather than silent diagrammatic steps.

9.4 Betweenness and Collinearity

The extensions of the two equal sides are represented by explicit betweenness hypotheses. These yield the collinearity and ray information needed to identify the auxiliary angles with the angles occurring in the Euclidean statement.

10 Formalization Notes

Proposition I.5 marks an important transition in the reconstruction of Book I.

The earlier propositions are centred on elementary constructions and a single application of the SAS theorem. Beginning with Proposition I.5, most subsequent propositions of Book I are expected to be reconstructed by composing previously established interface theorems.

This modular organization differs substantially from the presentation of the *Elements*. Euclid develops each proposition as an independent geometric argument, whereas the present reconstruction emphasizes reusable formal components. The resulting proofs are shorter, their logical dependencies are explicit, and each proposition becomes a thin verification layer built upon the synthetic theory already developed in the Hilbert interface.

11 Dependency Summary

The classical and formal architectures are sharply different:

```
Classical:
I.3 + two uses of I.4 + angle subtraction  ->  I.5

Formal:
hilbert_isosceles_base_angles
          +
hilbert_adjacent_angles_congruent         ->  euclid_proposition_5
```

The Book I theorem is therefore a thin interface layer over results already proved in the Hilbert development.

12 Conclusion

Proposition I.5 is the first substantial theorem about the internal symmetry of a triangle.

Starting from two congruent sides, Euclid constructs an extended auxiliary configuration and uses SAS twice to prove congruence of the base angles. The theorem also establishes the corresponding congruence of the exterior angles below the base.

The formal reconstruction shows that the geometric idea remains elementary, but its successful reuse depends on a well-developed interface for segment extension, layoff, betweenness, collinearity, and angle-congruence transport.

I.5 is structurally important far beyond the immediate statement. Its converse is Proposition I.6; the pair then supplies the isosceles machinery used in the uniqueness argument of I.7 and ultimately in the SSS theorem of I.8.

Thus Proposition I.5 begins a coherent four-proposition development

$$\mathrm{I.5} \longrightarrow \mathrm{I.6} \longrightarrow \mathrm{I.7} \longrightarrow \mathrm{I.8},$$

moving from isosceles symmetry to general triangle rigidity.